C

EDEXCEL INTERNATIONAL GCSE MATH (4MA1) • UNIT 1.01

Number, Surds, Bounds & Set Notation

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

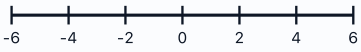
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	**Q2** B	**Q3** C	**Q4** D	**Q5** A
Q6 B	**Q7** C	**Q8** D	**Q9** D	**Q10** B
Q11 B	**Q12** C	**Q13** A	**Q14** C	**Q15** A
Q16 C	**Q17** D	**Q18** B	**Q19** C	**Q20** B
Q21 C	**Q22** C	**Q23** C	**Q24** B	**Q25** A

Number, Surds, Bounds & Set Notation Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 6x + 20$ in the domain of Number, Surds, Bounds & Set Theory. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(5)$.



A 26 ✓

B 29

C 15

D 24

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Number, Surds, Bounds & Set Theory):

1. Complete the square: $x^2 - 6x + 20 = (x-3)^2 - 9 + 20 = (x-3)^2 + 11$, giving minimum value $k = 11$. **[M1]**
2. Evaluate at $x = 5$: $f_1(5) = (2)^2 + 11 = 15$. **[M1]**
3. Sum the two quantities: $k + f_1(5) = 11 + 15 = 26$. **[A1]**

- **Final Verified Answer:** 26.

2 MCQ

A mathematical model in Number, Surds, Bounds & Set Theory is defined by the cubic function $y_2(x) = 2x^3 - 3x^2 + 15x - 8$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 69

B 51 ✓

C 36

D 27

WORKED SOLUTION & VERIFICATION

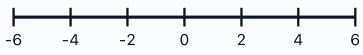
Mark Scheme & Examiner Solution (Number, Surds, Bounds & Set Theory):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 6x^2 - 6x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 6(9) - 6(3) + 15 = 51$. **[M1][A1]**

- **Final Verified Answer:** 51.

3 MCQ

In Number, Surds, Bounds & Set Theory, two displacement vectors are given by $\mathbf{u}_3 = \langle 3, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



A -29

B 73

C 44 ✓

D 156

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Number, Surds, Bounds & Set Theory):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (3)(9) + (-8)(7) = 27 + (-56) = -29$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 3^2 + (-8)^2 = 73$. **[M1]**
3. Combine: $-29 + 73 = 44$. **[A1]**

- **Final Verified Answer:** 44.

4 MCQ

The linear growth profile shown in the diagram for Number, Surds, Bounds & Set Theory follows an arithmetic progression with first term $a_1 = 4$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 88

B 780

C 1320

D 690 ✓

WORKED SOLUTION & VERIFICATION

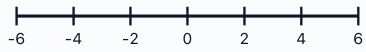
Mark Scheme & Examiner Solution (Number, Surds, Bounds & Set Theory):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 4, d = 6$: $S_{15} = \frac{15}{2}[8 + 14 \times 6] = \frac{15}{2} \times 92 = 690$. **[M1][A1]**

- **Final Verified Answer:** 690.

5 MCQ

In Number, Surds, Bounds & Set Theory, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 12x)\,\mathrm{d}x$.



- **A** 72 ✓
- **B** 96
- **C** 84
- **D** 37

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Number, Surds, Bounds & Set Theory):

1. Integrate term by term: $\int (18x^2 + 12x)\,\mathrm{d}x = [6x^3 + 6x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 6(2)^2) - 0 = 48 + 24 = 72$. **[M1][A1]**

- **Final Verified Answer:** 72.

6 MCQ

In Number, Surds, Bounds & Set Theory, consider the quadratic function $f_6(x) = x^2 - 8x + 99$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(11)$.

- **A** 219
- **B** 215 ✓
- **C** 132
- **D** 208

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Number, Surds, Bounds & Set Theory):

1. Complete the square: $x^2 - 8x + 99 = (x - 4)^2 - 16 + 99 = (x - 4)^2 + 83$, giving minimum value $k = 83$. **[M1]**
2. Evaluate at $x = 11$: $f_6(11) = (7)^2 + 83 = 132$. **[M1]**
3. Sum the two quantities: $k + f_6(11) = 83 + 132 = 215$. **[A1]**

- **Final Verified Answer:** 215.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 2x^3 - 4x^2 + 33x - 15$ studied in Number, Surds, Bounds & Set Theory. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

- **A** 417
- **B** 320
- **C** 353 ✓
- **D** 73

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Number, Surds, Bounds & Set Theory):

1. Differentiate term by term using the power rule: $\frac{\mathrm{d}y}{\mathrm{d}x} = 6x^2 - 8x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=8} = 6(64) - 8(8) + 33 = 353$. **[M1][A1]**

- **Final Verified Answer:** 353.

8 MCQ

In Number, Surds, Bounds & Set Theory, two displacement vectors are given by $\mathbf{u}_8 = \langle 4, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A -3

B 65

C 132

D 62 ✓

WORKED SOLUTION & VERIFICATION

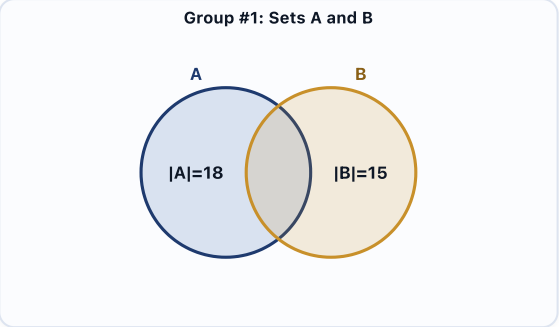
Mark Scheme & Examiner Solution (Number, Surds, Bounds & Set Theory):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (4)(8) + (-7)(5) = 32 + (-35) = -3$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 4^2 + (-7)^2 = 65$. **[M1]**
3. Combine: $-3 + 65 = 62$. **[A1]**

- **Final Verified Answer:** 62.

9 MCQ

In a survey of students in Year 11 Group #1, $|A| = 18$ study French, $|B| = 15$ study Spanish, $|A \cap B| = 5$ study both languages, and 4 study neither. Find the total number of students $|E|$.



A 37

B 14

C 28

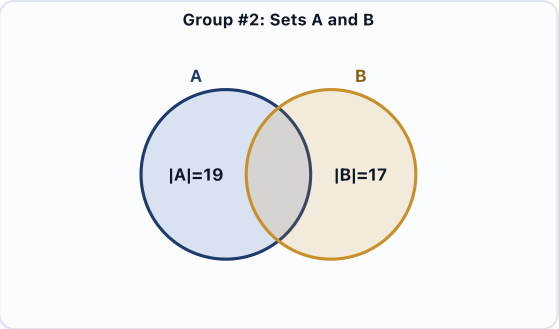
D 32 ✓

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 18 + 15 - 5 = 28$. Adding the 4 outside $A \cup B$ gives $|E| = 32$.

10 MCQ

In a survey of students in Year 11 Group #2, $|A| = 19$ study French, $|B| = 17$ study Spanish, $|A \cap B| = 6$ study both languages, and 5 study neither. Find the total number of students $|E|$.



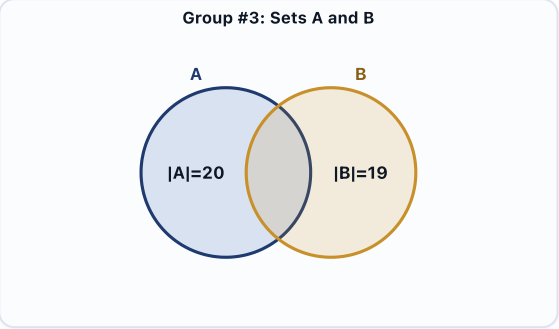
A	14	B	35 ✓
C	41	D	30

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 19 + 17 - 6 = 30$. Adding the 5 outside $A \cup B$ gives $|E| = 35$.

11 MCQ

In a survey of students in Year 11 Group #3, $|A| = 20$ study French, $|B| = 19$ study Spanish, $|A \cap B| = 7$ study both languages, and 6 study neither. Find the total number of students $|E|$.



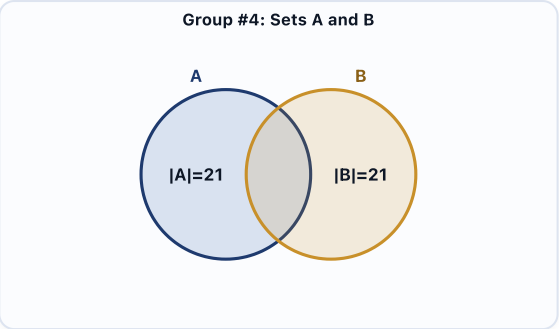
A	32	B	38 ✓
C	45	D	14

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 20 + 19 - 7 = 32$. Adding the 6 outside $A \cup B$ gives $|E| = 38$.

12 MCQ

In a survey of students in Year 11 Group #4, $|A| = 21$ study French, $|B| = 21$ study Spanish, $|A \cap B| = 8$ study both languages, and 7 study neither. Find the total number of students $|\mathrm{E}|$.



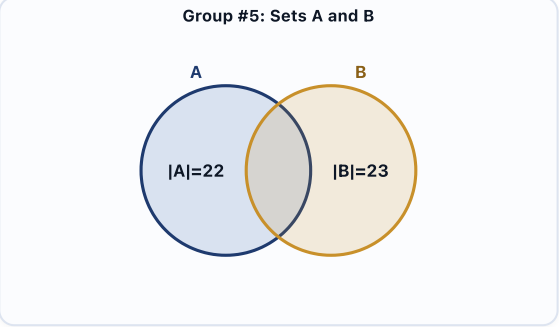
A 49	B 14
C 41 ✓	D 34

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 21 + 21 - 8 = 34$. Adding the 7 outside $A \cup B$ gives $|\mathrm{E}| = 41$.

13 MCQ

In a survey of students in Year 11 Group #5, $|A| = 22$ study French, $|B| = 23$ study Spanish, $|A \cap B| = 9$ study both languages, and 8 study neither. Find the total number of students $|\mathrm{E}|$.



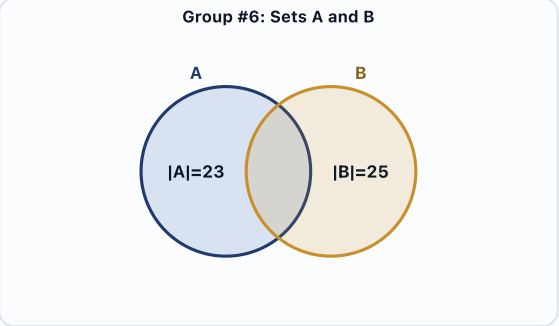
A 44 ✓	B 36
C 53	D 14

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 22 + 23 - 9 = 36$. Adding the 8 outside $A \cup B$ gives $|\mathrm{E}| = 44$.

14 MCQ

In a survey of students in Year 11 Group #6, $|A| = 23$ study French, $|B| = 25$ study Spanish, $|A \cap B| = 10$ study both languages, and 9 study neither. Find the total number of students $|E|$.



A 14

B 57

C 47 ✓

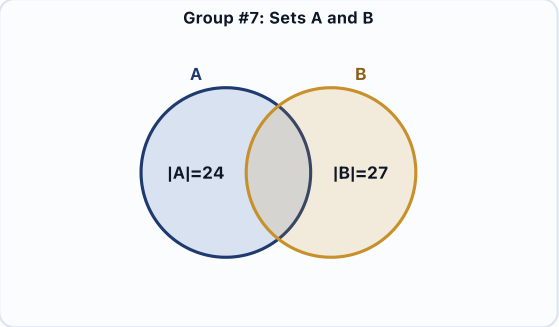
D 38

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 23 + 25 - 10 = 38$. Adding the 9 outside $A \cup B$ gives $|E| = 47$.

15 MCQ

In a survey of students in Year 11 Group #7, $|A| = 24$ study French, $|B| = 27$ study Spanish, $|A \cap B| = 11$ study both languages, and 10 study neither. Find the total number of students $|E|$.



A 50 ✓

B 61

C 40

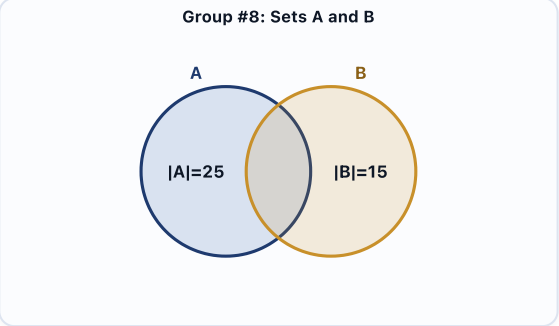
D 14

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 24 + 27 - 11 = 40$. Adding the 10 outside $A \cup B$ gives $|E| = 50$.

16 MCQ

In a survey of students in Year 11 Group #8, $|A| = 25$ study French, $|B| = 15$ study Spanish, $|A \cap B| = 12$ study both languages, and 4 study neither. Find the total number of students $|E|$.



A 44

B 28

C 32 ✓

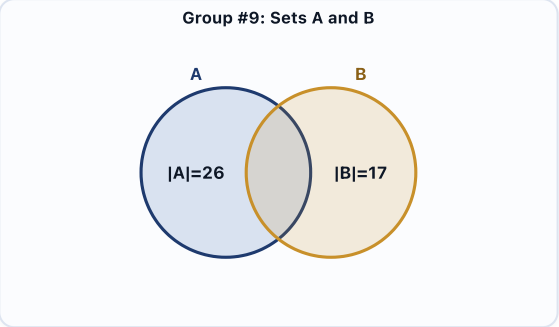
D 14

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 25 + 15 - 12 = 28$. Adding the 4 outside $A \cup B$ gives $|E| = 32$.

17 MCQ

In a survey of students in Year 11 Group #9, $|A| = 26$ study French, $|B| = 17$ study Spanish, $|A \cap B| = 5$ study both languages, and 5 study neither. Find the total number of students $|E|$.



A 48

B 38

C 14

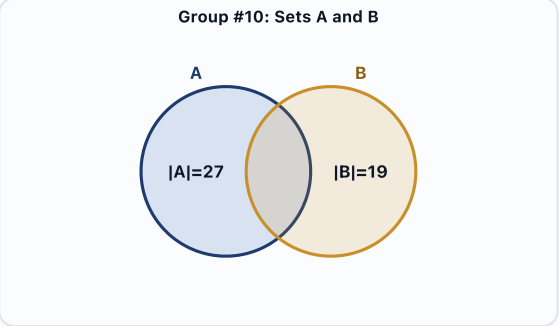
D 43 ✓

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 26 + 17 - 5 = 38$. Adding the 5 outside $A \cup B$ gives $|E| = 43$.

18 MCQ

In a survey of students in Year 11 Group #10, $|A| = 27$ study French, $|B| = 19$ study Spanish, $|A \cap B| = 6$ study both languages, and 6 study neither. Find the total number of students $|E|$.



A 14

B 46 ✓

C 40

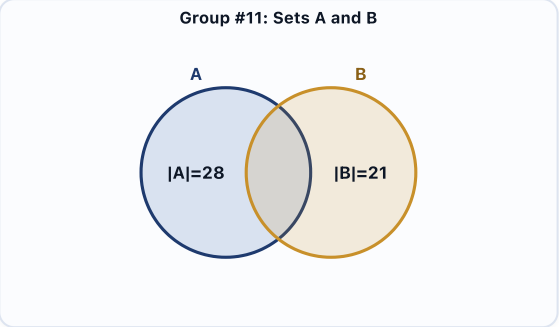
D 52

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 27 + 19 - 6 = 40$. Adding the 6 outside $A \cup B$ gives $|E| = 46$.

19 MCQ

In a survey of students in Year 11 Group #11, $|A| = 28$ study French, $|B| = 21$ study Spanish, $|A \cap B| = 7$ study both languages, and 7 study neither. Find the total number of students $|E|$.



A 42

B 56

C 49 ✓

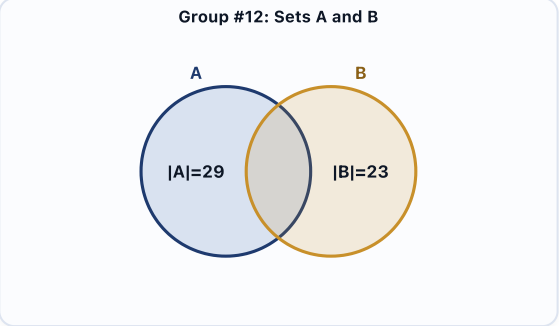
D 14

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 28 + 21 - 7 = 42$. Adding the 7 outside $A \cup B$ gives $|E| = 49$.

20 MCQ

In a survey of students in Year 11 Group #12, $|A| = 29$ study French, $|B| = 23$ study Spanish, $|A \cap B| = 8$ study both languages, and 8 study neither. Find the total number of students $|E|$.



A 60

B 52 ✓

C 44

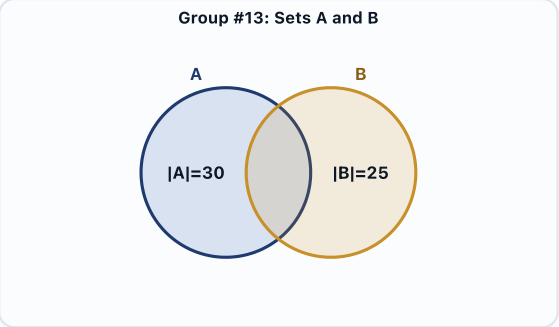
D 14

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 29 + 23 - 8 = 44$. Adding the 8 outside $A \cup B$ gives $|E| = 52$.

21 MCQ

In a survey of students in Year 11 Group #13, $|A| = 30$ study French, $|B| = 25$ study Spanish, $|A \cap B| = 9$ study both languages, and 9 study neither. Find the total number of students $|E|$.



A 64

B 14

C 55 ✓

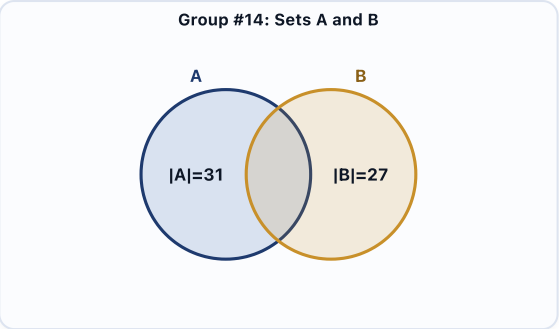
D 46

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 30 + 25 - 9 = 46$. Adding the 9 outside $A \cup B$ gives $|E| = 55$.

22 MCQ

In a survey of students in Year 11 Group #14, $|A| = 31$ study French, $|B| = 27$ study Spanish, $|A \cap B| = 10$ study both languages, and 10 study neither. Find the total number of students $|E|$.



A 48

B 14

C 58 ✓

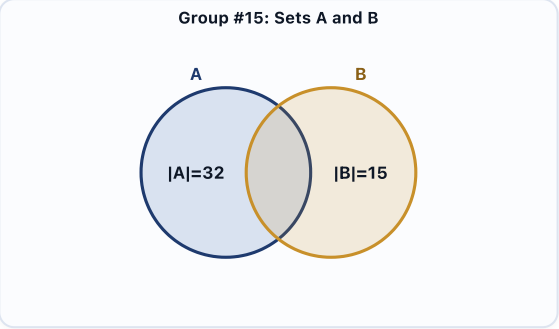
D 68

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 31 + 27 - 10 = 48$. Adding the 10 outside $A \cup B$ gives $|E| = 58$.

23 MCQ

In a survey of students in Year 11 Group #15, $|A| = 32$ study French, $|B| = 15$ study Spanish, $|A \cap B| = 11$ study both languages, and 4 study neither. Find the total number of students $|E|$.



A 51

B 14

C 40 ✓

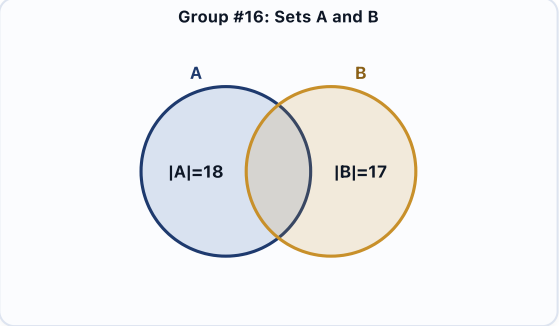
D 36

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 32 + 15 - 11 = 36$. Adding the 4 outside $A \cup B$ gives $|E| = 40$.

24 MCQ

In a survey of students in Year 11 Group #16, $|A| = 18$ study French, $|B| = 17$ study Spanish, $|A \cap B| = 12$ study both languages, and 5 study neither. Find the total number of students $|\mathrm{E}|$.



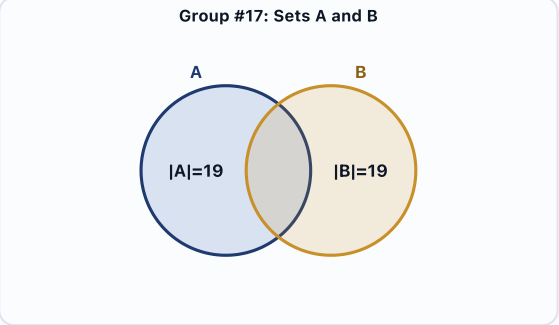
- **A** 14
- **B** 28 ✓
- **C** 23
- **D** 40

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 18 + 17 - 12 = 23$. Adding the 5 outside $A \cup B$ gives $|\mathrm{E}| = 28$.

25 MCQ

In a survey of students in Year 11 Group #17, $|A| = 19$ study French, $|B| = 19$ study Spanish, $|A \cap B| = 5$ study both languages, and 6 study neither. Find the total number of students $|\mathrm{E}|$.



- **A** 39 ✓
- **B** 44
- **C** 14
- **D** 33

WORKED SOLUTION & VERIFICATION

By inclusion-exclusion, $|A \cup B| = 19 + 19 - 5 = 33$. Adding the 6 outside $A \cup B$ gives $|\mathrm{E}| = 39$.